An Agmon bound for the Kogut–Susskind ground state: finite-volume global magnetic-energy tails

The global/local correction is incorporated below; the magnetic-energy identities note, §4, records the limitation of the proposed Gibbs-comparison repair (full-read).

> **Correction (2026-10-02).** The original title and lead claimed suppression at rate $1/g^2$ per locally excited plaquette and said this supplied the Hamiltonian large-field input for decimation. Those claims are **withdrawn**. Section 2 estimated one relaxation path, which can give only an upper bound on an infimum, whereas decay requires a lower bound for every path. It is replaced below by the global-excess bound below, with its volume dependence retained. The original pointwise estimate for $|\Omega(U)|^2$ is replaced by an integrated probability estimate. Sections 3–5 no longer assert local exponential-moment control or a successful match to the flow Jacobian. Lemma 1 and Corollary 2 survive; their application now states the weight margin and prefactor explicitly. The threshold e_0/B' is not the mean magnetic energy. Excited-state extensions use $|\psi|^2$ and do not assume positivity. This paragraph records the correction rather than silently removing the earlier claims.

On a fixed finite spatial lattice, Agmon's weighted eigenfunction identity gives a tail bound for the **total** magnetic potential of the Kogut–Susskind ground state. With

$$H = \frac{\hbar c}{a}\left[A'(-\Delta) + B'V\right], \qquad A' = \frac{g^2}{2}, \quad B' = \frac{2}{g^2}, \quad V = \sum_p \left(N - \operatorname{Re}\operatorname{tr} U_p\right),$$

write $e_0 = aE_0/(\hbar c)$ and $\bar{v} = e_0/B'$. If $M \geq \|\nabla V\|_\infty$ and $M > 0$, every path to $\{V \leq \bar{v}\}$ has Agmon length at least

$$\frac{4}{3g^2 M}\,(V - \bar{v})_+^{3/2}.$$

Consequently, for $0 < \delta < 1$ and $s > 0$,

$$\mathbb{P}_\Omega(V - \bar{v} \geq s) \leq C_{\epsilon,\delta} \exp\left[-\frac{8(1-\delta)}{3g^2 M} s^{3/2}\right],$$

with a finite-volume prefactor specified in §2. On a periodic three-dimensional spatial lattice one may take $M = 2\sqrt{2N|\mathcal{E}|}$ and $|\mathcal{E}| = |\mathcal{P}|$. A total excess proportional to volume then has an extensive exponent; a fixed total excess has an exponent that vanishes as the volume grows. No volume-uniform local large-field estimate, spectral gap or continuum claim follows. The written proofs below establish these finite-lattice statements without the original relaxation-path heuristic.

1. Setting and the weighted identity

Let $G = SU(N)$, $N \geq 2$, $g > 0$, $a > 0$, and let $\mathcal{M} = G^{\mathcal{E}}$ carry the product bi-invariant metric with Lie algebra normalization $\operatorname{tr}(T^a T^b) = \delta_{ab}/2$. Use a finite spatial lattice whose elementary plaquettes have four distinct links; the periodic cubic examples have at least three sites in each direction. Haar measure $d\mu$ is normalized. The operator $-\Delta$ is nonnegative. Its smooth bounded potential $V \geq 0$ gives a smooth normalized positive ground state Ω on this connected closed manifold, as in T1 of the obligations map (setting passage). The unique positive ground state is gauge invariant. In dimensionless units its equation is

$$-A'\Delta\Omega + (B'V - e_0)\Omega = 0, \qquad \int \Omega^2 d\mu = 1.$$

Lemma 1 (Agmon identity). For every real Lipschitz weight ρ,

$$A' \int_{\mathcal{M}} |\nabla(e^{\rho}\Omega)|^2 d\mu + \int_{\mathcal{M}} (B'V - e_0 - A'|\nabla\rho|^2) e^{2\rho}\Omega^2 d\mu = 0.$$

Proof. Multiply the eigenvalue equation by $e^{2\rho}\Omega$ and integrate by parts on the closed manifold. Almost everywhere,

$$|\nabla(e^{\rho}\Omega)|^2 = \nabla\Omega \cdot \nabla(e^{2\rho}\Omega) + e^{2\rho}\Omega^2|\nabla\rho|^2.$$

Substitution gives the identity; Lipschitz weights are valid weak test functions on this compact manifold. □

Corollary 2 (forbidden-region estimate). Put $q = B'V - e_0$, $\Sigma_\epsilon = \{q \geq \epsilon\}$, $\epsilon > 0$. If ρ is Lipschitz,

$$A'|\nabla\rho|^2 \leq q - \epsilon/2 \quad \text{on } \Sigma_\epsilon, \qquad \rho \leq R \quad \text{on } \mathcal{M} \setminus \Sigma_\epsilon,$$

then

$$\frac{\epsilon}{2} \int_{\Sigma_\epsilon} e^{2\rho}\Omega^2 d\mu \leq K_\rho e^{2R} \int_{\mathcal{M}\setminus\Sigma_\epsilon} \Omega^2 d\mu, \qquad K_\rho = e_0 + A'\|\nabla\rho\|_\infty^2.$$

More generally, replacing $\epsilon/2$ in the admissibility inequality by any $\eta > 0$ replaces the left coefficient by η. Thus

$$\int_{\mathcal{M}} e^{2\rho}\Omega^2 d\mu \leq e^{2R}\left(1 + \frac{K_\rho}{\eta}\right).$$

Proof. Drop the nonnegative gradient integral in Lemma 1. On the forbidden set the remaining coefficient is at least η; on its complement it is at least $-K_\rho$ because $V \geq 0$. Split the integral and use normalization. The complementary weighted mass is at most e^{2R}. □

The Agmon distance to $\mathcal{A} = \{q \leq 0\}$ is

$$d(U) = \inf_{\gamma:\, U\to\mathcal{A}} \int_\gamma \sqrt{q_+/A'}\, |d\gamma|.$$

The positive part is essential: the metric vanishes in the allowed region. This distance is Lipschitz, vanishes on $\mathcal{A}$, and satisfies $A'|\nabla d|^2 \leq q_+$ almost everywhere. Indeed appending a short segment bounds its increment by the local weighted length; the bounded weight also bounds its ordinary Lipschitz constant. Among weights vanishing on $\mathcal{A}$ with this gradient constraint, d is maximal, by integrating the constraint along every path. This zero-set and gradient constraint is distinct from Corollary 2's strict margin condition.

For $\rho = (1-\delta)d$, define $\theta_\delta = 1 - (1-\delta)^2 = 2\delta - \delta^2 > 0$. On Σ_ϵ,

$$q - A'|\nabla\rho|^2 \geq \theta_\delta q \geq \theta_\delta \epsilon.$$

The original $\epsilon/2$ margin is therefore guaranteed by this argument only if $\delta \geq 1 - 1/\sqrt{2}$. For arbitrary $0 < \delta < 1$, use the generalized margin $\eta = \theta_\delta\epsilon$ instead. Let $q_{\max} = \max_{\mathcal{M}} q_+$ and $R_{\epsilon,\delta} = (1-\delta)\sup_{\{q<\epsilon\}} d$. Then

$$\int e^{2(1-\delta)d}\Omega^2 d\mu \leq e^{2R_{\epsilon,\delta}}\left(1 + \frac{e_0 + (1-\delta)^2 q_{\max}}{\theta_\delta\epsilon}\right).$$

All constants are finite at fixed lattice and coupling. In particular,

$$R_{\epsilon,\delta} \leq (1-\delta)\,\mathrm{diam}(\mathcal{M})\sqrt{q_{\max}/A'},$$

since $\mathcal{A}$ is nonempty ($V = 0$ is attainable and $e_0 \geq 0$). This is not a uniform constant in volume or coupling. No pointwise bound on Ω follows just by dropping the gradient term in this identity.

2. A lower bound over all paths and a global tail

The threshold $\bar{v} = e_0/B'$ satisfies

$$\bar{v} = \langle V\rangle_\Omega + \frac{A'}{B'}\int |\nabla\Omega|^2 d\mu \geq \langle V\rangle_\Omega.$$

It is an energy threshold, not the ground-state mean magnetic energy. The constant trial function gives $e_0 \leq B'\langle V\rangle_{\mathrm{Haar}} = B'N|\mathcal{P}|$, since each plaquette has a Haar-distributed holonomy. Also $V \leq 2N|\mathcal{P}|$, so $q_{\max} \leq 2B'N|\mathcal{P}|$.

Proposition 3 (global excess, corrected). For any $M \geq \|\nabla V\|_\infty$, $M > 0$, set

$$f = V - \bar{v}, \qquad F(U) = \frac{4}{3g^2M} f(U)_+^{3/2}.$$

Then $d \geq F$. For any $\epsilon > 0$ and $0 < \delta < 1$ set

$$R_F = (1-\delta)\frac{4}{3g^2M}\left(\frac{\epsilon}{B'}\right)^{3/2}, \qquad C_{\epsilon,\delta} = e^{2R_F}\left(1 + \frac{e_0 + (1-\delta)^2 q_{\max}}{\theta_\delta \epsilon}\right).$$

For $\mathbb{P}_\Omega(B) = \int_B \Omega^2 d\mu$,

$$\mathbb{P}_\Omega(f \geq s) \leq \min\left\{1,\ C_{\epsilon,\delta}\exp\left[-\frac{8(1-\delta)}{3g^2M}s^{3/2}\right]\right\}, \qquad s > 0.$$

Proof. On any path from U with $f(U) > 0$ to $\{f \leq 0\}$, stop at its first zero of f. With arclength parameter r, $|f'(r)| \leq M$. The total variation of $\frac{2}{3}f^{3/2}$ along this segment is at least its endpoint difference, even if f is not monotone. Hence

$$\int_\gamma \sqrt{(B'/A')f_+}\, dr \geq \frac{\sqrt{B'/A'}}{M}\int \sqrt{f}\,|f'|\, dr \geq \frac{2\sqrt{B'/A'}}{3M} f(U)^{3/2} = F(U).$$

Taking the infimum proves the all-path lower bound. For an explicit prefactor, apply Corollary 2 directly to $\rho = (1-\delta)F$, rather than estimating d on the complementary set. Differentiation gives

$$A'|\nabla F|^2 = \frac{B' f_+}{M^2}|\nabla V|^2 \leq q_+.$$

Thus the margin on Σ_ϵ is $\theta_\delta\epsilon$, $\rho \leq R_F$ on its complement, and $K_\rho \leq e_0 + (1-\delta)^2 q_{\max}$. The full weighted mass is bounded by $C_{\epsilon,\delta}$. On $\{f \geq s\}$ the weight is at least $\exp[8(1-\delta)s^{3/2}/(3g^2M)]$, proving the tail. □

For the periodic three-dimensional spatial cubic lattice each link is in four plaquettes. The completeness relation yields $|\nabla_\ell \operatorname{Re}\operatorname{tr} U_p| \leq \sqrt{N/2}$ (the upper-bound note, Corollary 2, proof passage). The triangle inequality and sum of squared link components give

$$\|\nabla V\|_\infty \leq 4\sqrt{N/2}\sqrt{|\mathcal{E}|} = 2\sqrt{2N|\mathcal{E}|} =: M.$$

In other spatial dimensions or geometries the plaquette incidence must be changed; this constant is specifically for three spatial dimensions. With $P = |\mathcal{P}| = |\mathcal{E}|$ and $s = wP$, Proposition 3 becomes

$$\mathbb{P}_\Omega(f \geq wP) \leq \min\left\{1,\ C_{\epsilon,\delta}\exp\left[-\frac{2\sqrt{2}(1-\delta)}{3}\frac{w^{3/2}}{g^2\sqrt{N}}P\right]\right\}.$$

At fixed g, N, ϵ, δ, $R_F = O(P^{-1/2})$ and the displayed prefactor is at most $O(P)$. Thus this is a genuine global extensive-tail bound for fixed $w > 0$. When w is a fixed positive multiple of N, the exponent per plaquette is of order N/g^2, correcting the earlier $\sqrt{N}/g^2$ scaling. Coupling dependence of the prefactor must still be retained in any weak-coupling limit.

For a fixed *total* excess $s = nv$ the exponent instead scales as

$$(nv)^{3/2}/(g^2\sqrt{NP}),$$

which vanishes as $P \to \infty$. A local event need not imply even this total excess, since the complementary plaquettes can lower V. The argument supplies no volume-uniform local estimate.

3. Comparison with the Euclidean weight

The Wilson action is

$$S_w = \frac{2N}{g_E^2} \sum_p \left(1 - \frac{1}{N} \operatorname{Re} \operatorname{tr} U_p\right) = \frac{2}{g_E^2} V.$$

An increase $\Delta V = nv$ multiplies its **unnormalized density** by e^{-2nv/g_E^2}. This is an exact action-cost statement. Proposition 3 instead bounds an integrated ground-state probability of a global excess with exponent proportional to $s^{3/2}/(g^2 M)$ and the displayed prefactor. Both expressions contain an inverse-square coupling factor; this comparison neither identifies g_E with g in a continuum limit nor proves equal local probabilities. Plaquettes share links, so a product of plaquette action factors is not a product measure of independent plaquette variables. Normalization and integration require additional estimates even on the Euclidean side.

The original assertion that local large-field regions are exponentially rare at the required rate in the Hamiltonian ground state is withdrawn. The global estimate does not supply the local input sought in the typical-field note or the flow-before-decimation note.

4. Scope, excited states and the missing local input

The surviving statements are the weighted identity, the forbidden-set estimate and the global integrated tail. At finite volume all exponential moments of a bounded local plaquette potential are already finite by compactness, for every finite rate. That fact supplies no uniform control as the volume, cutoff or observable changes; the original claim of useful local moments below a rate $\lambda = 2\sqrt{2v}/g^2$ is withdrawn.

For a normalized excited eigenfunction ψ_j with dimensionless energy e_j, positivity is unavailable: real excited eigenfunctions orthogonal to Ω change sign. Nevertheless the identity holds as

$$A' \int |\nabla(e^{\rho}\psi_j)|^2 d\mu + \int (B'V - e_j - A'|\nabla\rho|^2) e^{2\rho} |\psi_j|^2 d\mu = 0.$$

For real eigenfunctions the proof is unchanged; for complex ones test with $e^{2\rho}\overline{\psi_j}$ and take the real part. Corollary 2 then applies with e_j and $|\psi_j|^2$, with its corresponding global threshold and constants. It does not require taking $\log \psi_j$ or assuming a positive excited state.

A superposition of eigenfunctions is not itself an eigenfunction. For a whole finite-volume spectral subspace below e_* one can choose a common admissible weight relative to $B'V - e_*$ and bound its weighted norm by summing the individual estimates: if its rank is m and each weighted eigenfunction norm squared is at most C_*, pointwise Cauchy–Schwarz gives a bound mC_* for any normalized superposition. Compactness gives finite rank at fixed e_*. Neither the rank nor the constants have been controlled uniformly in volume or cutoff here. Merely replacing e_0 by the first excited energy e_1 does not prove a uniform low-energy-subspace estimate.

No successful comparison with $e^{2t\|G\|_\infty}$, the flow-Jacobian growth factor, follows from these global bounds. A local state-relative estimate with the required uniformity remains a separate obligation. The Gibbs-comparison attempt in the magnetic-energy identities note, §4 (full-read), has a threshold and prefactor determined by global energy data; its failure to supply uniform local control is not a proof that every possible local estimate is impossible. The transfer-measure note, §§1–2 (passages), describes a different route: any Euclidean estimate used there must survive the temporal-continuum and vacuum limits with its constants controlled. No such local estimate is established here.

5. Consequence for STATE

Keep this note as a finite-lattice global-energy result: Lemma 1, Corollary 2 with explicit margins and prefactors, and Proposition 3's all-path lower bound and integrated tail. The original local per-plaquette rate, pointwise decay claim, local moment criterion and claim to supply

the Hamiltonian decimation input are withdrawn. This corrected argument controls reuse; a local energy or measure estimate uniform in volume and cutoff remains an independent requirement. Excited-state identities survive with $|\psi_j|^2$, but uniform spectral-subspace and refinement estimates do not follow from them. No mass gap or continuum construction is claimed.